12(a)	$x = 7$	**A1**
12(b)(i)	$E = mc^2$	**C1**
	$= 1.66 \times 10^{-27} \times (3.0 \times 10^8)^2$ $= 1.494 \times 10^{-10}$ J	**C1**
	division by 1.6×10^{-13} clear to give 934 MeV	**A1**
12(b)(ii)	$\Delta m = (235.123 + 1.00863) - (94.945 + 138.955 + 2 \times 1.00863 + 7 \times 5.49 \times 10^{-4})$ **or** $\Delta m = 235.123 - (94.945 + 138.955 + 1 \times 1.00863 + 7 \times 5.49 \times 10^{-4})$	**C1**
	$= 0.21053$ u	**C1**
	energy $= 0.21053 \times 934$ $= 197$ MeV	**A1**
12(c)	kinetic energy of nuclei/particles/products/fragments	**B1**
	γ–ray photon energy	**B1**